

SPACE INSTRUMENTATION LAB

SPA617

Radio Astronomy Lab

RF Transmission, reflections and standing waves

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1 Aim

To understand signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies, and to determine important cable parameters experimentally.

2 Objectives

1. To determine the phase velocity and velocity factor of a signal in a coaxial cable from the frequency spacing between consecutive maxima or minima produced by reflection from an open-ended cable.
2. To study signal reflection from an open-ended cable and from a cable terminated with a 50 Ω load.
3. To estimate the magnitude of the reflection coefficient and verify that the coaxial cable is approximately matched to a 50 Ω termination, indicating that its characteristic impedance is close to 50 Ω.
4. To calculate the inductance and capacitance per unit length using the measured phase velocity and the approximate value $Z_0 = 50\Omega$.
5. To measure the insertion loss and attenuation per unit length of the coaxial cable.

3 Apparatus Required

RF signal generator, Digital storage oscilloscope (DSO), Spectrum analyser, RF Tee connector, Coaxial cable under test, Short reference coaxial cable, 50Ω termination, Open-circuit termination, obtained by leaving the cable end unconnected, Suitable RF adapters and connectors

4 Theory

4.1 Transmission lines and reflections

A transmission line is a structure that guides electromagnetic energy from one point to another. A coaxial cable supports transverse electromagnetic (TEM) propagation, for which both the electric and magnetic fields are transverse to the direction of propagation. The signal velocity in a cable is lower than the speed of light in vacuum because of the dielectric material between the conductors.

When a travelling wave encounters a load impedance Z_L that is different from the characteristic impedance Z_0 of the cable, a part of the wave is reflected toward the source. At any fixed observation point, the incident and reflected waves superimpose. Depending on their relative phase, the measured amplitude may be large because of constructive interference or small because of destructive interference.

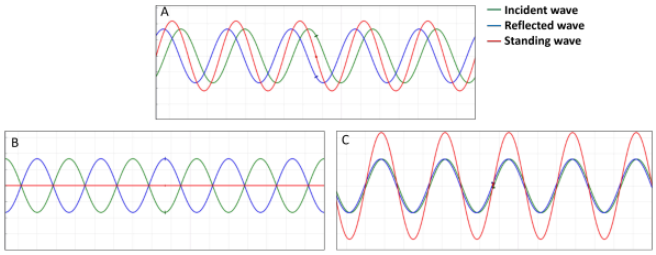


Figure 1: Superposition of incident and reflected waves giving constructive (maximum) and destructive (minimum) interference.

In this experiment, the oscilloscope measures the voltage at one fixed point, namely the Tee junction. It does not directly display the spatial standing-wave pattern along the cable. When the signal frequency is varied, the round-trip phase of the reflected wave changes, causing the measured amplitude at the Tee junction to vary periodically.

4.2 Phase velocity measurement

Let the physical length of the coaxial cable be ℓ and the phase velocity in the cable be v_p. The reflected signal travels a round-trip distance 2ℓ, so the round-trip time delay is

$$\tau = \frac{2\ell}{v_p}.$$

The round-trip phase of the reflected signal is

$$\phi_{RT} = 2\pi f\tau + \phi_\Gamma = \frac{4\pi f\ell}{v_p} + \phi_\Gamma,$$

where ϕ_Γ is the phase shift introduced by the termination. For an ideal open circuit, $\phi_\Gamma = 0$.

Two consecutive maxima, or two consecutive minima, are separated by a round-trip phase change of 2π. Therefore,

$$2\pi\,\Delta f\,\tau = 2\pi,$$

which gives

$$\tau = \frac{1}{\Delta f} \qquad \text{and} \qquad v_p = 2\ell\,\Delta f. \tag{1}$$

The velocity factor of the cable is

$$\mathrm{VF} = \frac{v_p}{c},$$

where $c \approx 3 \times 10^8$ m/s is the speed of light in vacuum. Here, Δf is the frequency difference between two consecutive maxima or two consecutive minima.

4.3 Reflection coefficient and characteristic impedance

The voltage reflection coefficient at the load is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0},$$

where Z_L is the load impedance and Z_0 is the characteristic impedance of the cable. For the important special cases:

- Open circuit: $Z_L \to \infty$ and $\Gamma = +1$.
- Short circuit: $Z_L = 0$ and $\Gamma = -1$.
- Matched load: $Z_L = Z_0$ and $\Gamma = 0$.

During a frequency sweep, the oscilloscope displays the superposition of the incident and reflected waves:

$$V_{osc} = V_{incident} + V_{reflected}.$$

When the incident and reflected waves are in phase, they interfere constructively and the measured amplitude is maximum:

$$V_{max} = |V_{incident}| + |V_{reflected}|.$$

When they are out of phase by 180°, they interfere destructively and the measured amplitude is minimum. Assuming the incident-wave amplitude is greater than or equal to the reflected-wave amplitude,

$$V_{min} = |V_{incident}| - |V_{reflected}|.$$

Adding and subtracting these two equations gives

$$|V_{incident}| = \frac{V_{max} + V_{min}}{2} \qquad \text{and} \qquad |V_{reflected}| = \frac{V_{max} - V_{min}}{2}.$$

Therefore, the magnitude of the reflection coefficient observed at the Tee junction can be estimated as

$$|\Gamma| = \frac{|V_{reflected}|}{|V_{incident}|} = \frac{V_{max} - V_{min}}{V_{max} + V_{min}}.$$

In general, the characteristic impedance of a transmission line can be estimated by connecting different known terminations, such as 25 Ω, 50 Ω, 100 Ω, an open circuit, and

a short circuit, and measuring the corresponding reflected-signal amplitudes. In this experiment, only a $50\,\Omega$ termination is used. The reflected signal obtained with this termination is compared with that obtained when the cable is left open. A significant reduction in the reflected-signal amplitude verifies that the cable is approximately matched to the $50\,\Omega$ termination and indicates that its characteristic impedance is close to $50\,\Omega$.

4.4 Inductance and capacitance per unit length

Let L' denote the inductance per unit length in H/m and C' denote the capacitance per unit length in F/m. For a low-loss transmission line,

$$v_p = \frac{1}{\sqrt{L'C'}} \qquad \text{and} \qquad Z_0 = \sqrt{\frac{L'}{C'}}.$$

Therefore,

$$L' = \frac{Z_0}{v_p} \qquad \text{and} \qquad C' = \frac{1}{Z_0 v_p}. \tag{2}$$

4.5 Cable attenuation

Cable attenuation is the reduction in signal level as the wave propagates through the cable. In this experiment, power is measured in dBm using a spectrum analyser. The insertion loss of the cable is

$$\mathrm{IL} = P_{ref} - P_{cable},$$

where P_{ref} is the power measured using a short reference connection and P_{cable} is the power measured after inserting the cable under test. The attenuation per unit length is then

$$\alpha_{\mathrm{dB}} = \frac{\mathrm{IL}}{\ell} \ \ \mathrm{dB/m}.$$

5 Procedure

Part A: Phase velocity and velocity factor

1. Measure the physical length ℓ of the cable.
2. Connect the signal generator, Tee connector, cable and oscilloscope as in Fig. 2, leaving the far end of the cable open. Set the oscilloscope input impedance to $1\,\mathrm{M}\Omega$.
3. Apply a sinusoidal signal of constant amplitude and sweep the frequency slowly over a range containing at least two maxima and two minima, recording the amplitude at suitable intervals.
4. Plot amplitude versus frequency, find f_1 and f_2 of consecutive maxima (or minima), and compute Δf, $v_p = 2\ell\Delta f$ and VF.

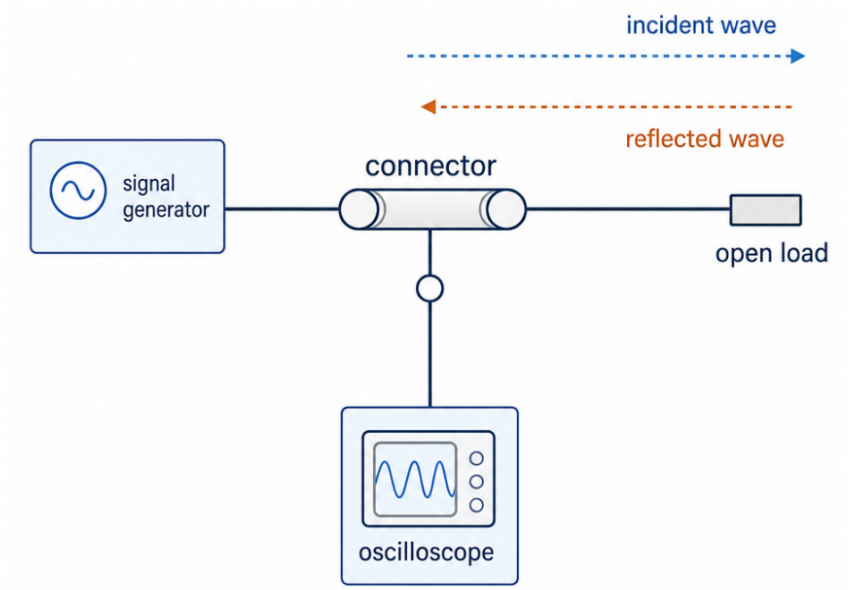


Figure 2: Experimental setup for the reflection measurement.

Part B: Reflection coefficient

1. Using the open-circuit data of Part A, read $V_{\max,\text{open}}$ and $V_{\min,\text{open}}$ over one ripple period and compute $|\Gamma_{\text{open}}|$.
2. Connect the $50\,\Omega$ termination, sweep the same frequency range at the same generator amplitude, and record $V_{\max,50}$ and $V_{\min,50}$.
3. Compute $|\Gamma_{50}|$ and compare it with $|\Gamma_{\text{open}}|$.

Part C: Inductance and capacitance

Calculate L' and C' from the measured v_p and $Z_0 \approx 50\,\Omega$.

Part D: Attenuation

1. Connect the generator to the spectrum analyser through the short reference cable at a fixed frequency and power; record P_{ref}.
2. Replace the reference cable by the cable under test, keeping everything else unchanged, and record P_{cable}.
3. Calculate IL and α_{dB}; repeat at other frequencies if required.

6 Observations and Calculations

Cable length: 0.92 m Generator amplitude: 1 V

Part A: RMS versus frequency (open end)

f (MHz)	RMS (mV)
20	1570
58	67 (min)
68	630
78	1200
88	1560
98	1730
108	1740 (max)
118	1640
138	1210
158	640
173	97 (min)
178	350
198	1280
210	1410
218	1280

Consecutive extrema (minima): $f_1 = 58$ MHz, $f_2 = 173$ MHz, $\Delta f = 115$ MHz.

From equation 1, we can calculate the phase velocity as:

$$v_p = 2 \times 0.92\,\text{m} \times 115 \times 10^6\,\text{Hz} = 2.116 \times 10^8\,\text{m/s}$$

and velocity factor (VF) as:

$$\text{VF} = \frac{2.116 \times 10^8}{3 \times 10^8} = 0.7053$$

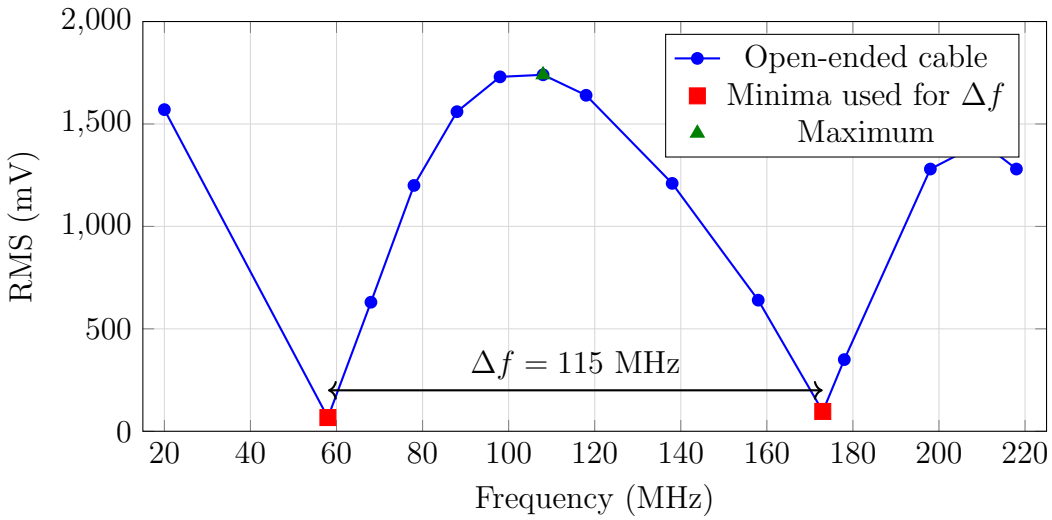


Figure 3: RMS voltage at the Tee junction versus frequency for the open-ended cable.

Part B: Reflection coefficient

50 Ω load connected: frequency sweep

f (MHz)	RMS (mV)
20	973.26
25	1040
58	954.19
68	982.07
78	972.11
88	973.04
98	990.06 (max)
108	982
118	950
128	942
138	926
148	900 (min)
158	904
168	933
173	938
178	950
198	929
210	812
218	731
238	691

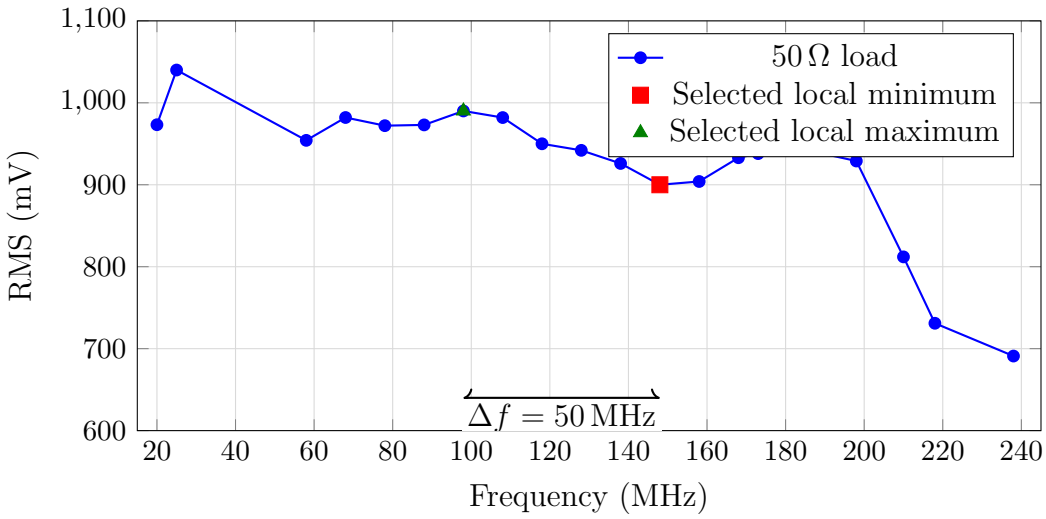


Figure 4: RMS voltage at the Tee junction versus frequency for the cable terminated with a 50 Ω load. The selected local extrema used for the reflection-coefficient estimate are highlighted.

Comparison of open circuit and 50 Ω load

The selected local maximum and minimum within the chosen ripple region are highlighted in Fig. 4. These are local extrema used for the reflection-coefficient estimate, not the global maximum and minimum across the full frequency sweep.

Termination	Selected $V_{\max}$ (mV)	Selected $V_{\min}$ (mV)	$\lvert\Gamma\rvert$
Open circuit	1740	97	0.894
50 Ω load	990.06	900	0.0477

The reflection coefficient is

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}},$$

which gives

$$|\Gamma_{\text{open}}| = \frac{1740 - 97}{1740 + 97} = 0.894, \qquad |\Gamma_{50}| = \frac{990.06 - 900}{990.06 + 900} = 0.0477.$$

The ripple falls by a factor of about 19 when the 50 Ω load replaces the open end, indicating that the cable is approximately matched to 50 Ω.

Part C: Inductance and capacitance

Using equation 2, we calculate the inductance and capacitance as:

$$L' = \frac{50}{2.116 \times 10^8} = 236.3\,\text{nH/m}, \qquad C' = \frac{1}{50 \times 2.116 \times 10^8} = 94.5\,\text{pF/m}$$

Part D: Attenuation

f (MHz)	P_{ref} (dBm)	P_{cable} (dBm)	IL (dB)	α_{dB} (dB/m)
50	0	-1.92	1.92	2.0867

Summary of results

Quantity	Symbol	Value
Cable length (m)	ℓ	0.92
Frequency spacing (MHz)	Δf	115
Phase velocity (m/s)	v_p	2.116×10^8
Velocity factor	v_p/c	0.7053
Approx. characteristic impedance (Ω)	Z_0	50
Inductance per unit length (H/m)	L'	236.3 nH/m
Capacitance per unit length (F/m)	C'	94.5 pF/m
Attenuation (dB/m)	α_{dB}	2.0867

7 Conclusions

The frequency dependence of the measured signal amplitude for the open-ended coaxial cable showed a ripple pattern caused by the interference of incident and reflected waves. The frequency spacing between consecutive minima was measured as $\Delta f = 115\,\text{MHz}$. Using the cable length of 0.92 m, the phase velocity was calculated to be $v_p = 2.116 \times 10^8\,\text{m/s}$, corresponding to a velocity factor of VF = 0.7053. The velocity factor being less than unity is consistent with electromagnetic wave propagation in a dielectric-filled coaxial cable.

The magnitude of the reflection coefficient was estimated to be $|\Gamma_{\text{open}}| = 0.894$ for the open-ended cable and $|\Gamma_{50}| = 0.0477$ for the cable terminated with a $50\,\Omega$ load, using the selected voltage extrema. The substantial reduction in the estimated reflection coefficient indicates that the $50\,\Omega$ termination considerably reduces reflections and that the cable is approximately matched to $50\,\Omega$.

Assuming a characteristic impedance of $Z_0 = 50\,\Omega$ and using the measured phase velocity, the inductance and capacitance per unit length were estimated as $L' = 236.3\,\text{nH/m}$ and $C' = 94.5\,\text{pF/m}$, respectively.

At 50 MHz, the reference and cable power measurements were 0 dBm and -1.92 dBm, respectively. The resulting insertion loss was 1.92 dB, giving a measured insertion loss per unit length of approximately 2.087 dB/m for the 0.92 m cable under the experimental setup. This value may also include connector losses and impedance-mismatch effects.

Possible sources of measurement uncertainty include the finite frequency step during the sweep, uncertainty in identifying the voltage extrema, cable-length measurement errors, connector and Tee-junction mismatches, non-ideal termination impedance, and frequency-dependent instrument response. Cable losses and reflections within the measurement setup may also affect the estimated reflection coefficients and attenuation.

8 Key Takeaway

The part I found most interesting was studying how reflections and interference affect the measured signal in a coaxial transmission line. I particularly liked observing how the RMS voltage varied with frequency for open-circuit and $50\,\Omega$-terminated conditions. The experiment helped me understand how impedance matching reduces reflections and improves signal transmission. Overall, this experiment connected theoretical concepts of wave propagation and interference with practical RF measurements.